

Study on reduction and carburization behaviors of iron phases for iron-based Fischer–Tropsch synthesis catalyst[☆]



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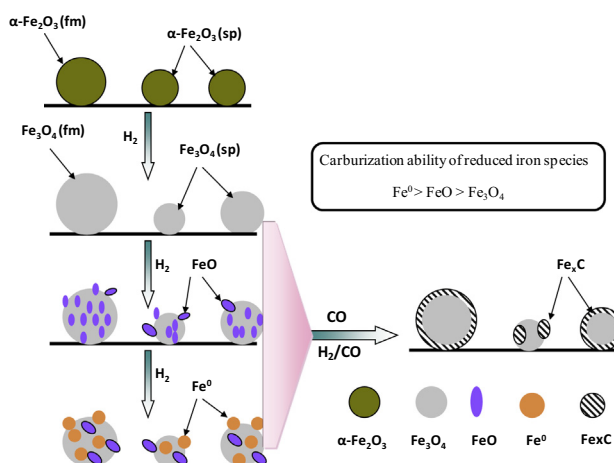
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HIGHLIGHTS

- Phase transformation during reduction is following as $\alpha\text{-Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4 \rightarrow \text{FeO} \rightarrow \alpha\text{-Fe}$.
- Carburization ability of reduced iron phases is following as $\alpha\text{-Fe} > \text{FeO} > \text{Fe}_3\text{O}_4$.
- Iron carbides are formed on the Fe(II) oxide species.
- Hydrocarbons species are formed gradually on the surface of iron carbides.

GRAPHICAL ABSTRACT



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ABSTRACT

Reduction and carburization behaviors of iron phases over a precipitated iron-based Fischer–Tropsch synthesis (FTS) catalyst were investigated by some techniques of Mössbauer effect spectroscopy (MES), X-ray photoelectron spectroscopy (XPS) and diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) as well as H_2 & CO temperature-programmed desorption (H_2 & CO -TPD). It was found that in H_2 atmosphere phase transformation of iron phases involved $\alpha\text{-Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4 \rightarrow \text{FeO} \rightarrow \alpha\text{-Fe}$, both occurring in the bulk and on the surface layers. All of reduced iron species took place the carburization reaction, whereas carburizing ability was following the order $\alpha\text{-Fe} > \text{FeO} > \text{Fe}_3\text{O}_4$. During FTS both iron carbides and Fe(II) oxide species reached a balance state without appearing the intermediate $\alpha\text{-Fe}$. The conversion of reduced iron phases to iron carbides (especially for $\chi\text{-Fe}_5\text{C}_2$) on the surface layers played a positive role in promoting the formation of hydrocarbons species.

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1. Introduction

Fischer–Tropsch synthesis (FTS) is a highly developed and technically proven route to transform coal, natural gas and lignocelluloses biomass into liquid alkanes, which are suitable for gasoline, diesel and jet fuel applications [1–3]. Due to higher water–gas shift activity, lower cost and favorable engineering characteristics, the

Fe-based catalysts are used widely in the commercial FTS processes with lower H_2/CO ratio syngas derived from coal or biomass [4,5]. However, the Fe catalyst undergoes the most complicated phase changes during FTS compared to other Fischer–Tropsch synthesis catalysts (including Ru, Ni and Co) [6–8].

Generally, the fresh Fe catalyst is consisted mainly of $\alpha-Fe_2O_3$, which may be converted to different reduced iron species (such as Fe_3O_4 , FeO or metallic iron) under H_2 atmosphere [9,10]. Furthermore, the reduced Fe_3O_4 is transformed continually to metastable FeO phase, or reduced directly to metallic iron [11]. It is generally accepted that the strong interaction of metal-support in the supported iron catalysts may stabilize the metastable FeO. Under CO or syngas atmosphere these reduced iron species could be converted to different types of iron carbides (such as $\chi-Fe_5C_2$, $\epsilon'-Fe_{2.2}C$, and $\theta-Fe_3C$). The results of Chen et al. [12] indicated that Fe_3O_4 was reduced firstly to metallic Fe, and then carburized to iron carbides, whereas Li et al. [13] reported that the reduced Fe_3O_4 was transformed directly to iron carbides without the formation of intermediate metallic iron during FTS. In addition, several iron phases including reduced iron species and iron carbides have been found to coexist or transform each other during reduction and carburization processes [14,15]. Bukur et al. [16] suggested that magnetite, metallic iron and iron carbides could be coexisted in the Fe-based catalysts under syngas atmosphere. The results of Zhang et al. [17] showed that there was only the appearance of Fe_3O_4 and iron carbides during FTS, without observing metallic iron. Due to complex changes of iron phases during reduction and carburization processes, reduction and carburization behaviors of different iron phases are not still understood clearly.

On the other hand, a large number of studies are devoted to investigate the adsorption properties of surface iron species [18–20]. Some compounds such as formats, hydrocarbons and carboxylates have been reported to be appeared on the catalyst surface, whereas these species are influenced obviously by the complex nature of iron phases [21]. The surface adsorption properties of iron-based catalysts supported on SiO_2 , Al_2O_3 and MgO have been widely investigated by infrared spectroscopy [22,23]. The surface species may be adsorbed on the support and metal particles, respectively, exhibiting different vibration properties. In addition, adsorption of CO and NO on unsupported iron catalysts has been extensively studied [24,25]. The results of Jiang et al. [25] indicated that both the Fe(III) and Fe(II) species were located on the surface of oxidized Fe sample. However, the adsorbed species may be different on the surface of different iron phases, and take place further changes with phase transformation during reduction and carburization, affecting ultimately the FTS performances. The reports on adsorption properties of different iron phases are very rare.

Therefore, the aim of this work is to investigate reduction and carburization behaviors as well as adsorption properties of different iron phases over a precipitated Fe–Mn FTS catalyst. The bulk and surface phase compositions as well as surface adsorbed species were characterized by Mössbauer effect spectroscopy (MES), X-ray photoelectron spectroscopy (XPS), hydrogen & CO temperature-programmed desorption (H_2 &CO-TPD) and diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS), respectively. This systematic approach provided a detailed observation of surface species and micro-compositions of the iron catalyst in different FTS stages.

2. Experimental

2.1. Catalyst preparation

A typical precipitated Fe–Mn–K– SiO_2 catalyst used in the present study was prepared by a combination of co-precipitation and

spray drying method. The detailed preparation method has been described elsewhere [26]. In brief, a solution containing $Fe(NO_3)_3$ and $Mn(NO_3)_2$ with the desired ratio was added into a continuously stirred tank containing de-ionized water at 80 °C. A NH_4OH solution was added simultaneously into the tank to maintain the pH value of 9.0. After precipitation, the precipitate was washed completely, and then filtered. Required amounts of silica sol solution, K_2CO_3 solution and de-ionized water were successively added into the filtrate, and the mixture was re-slurried, spray-dried and calcined at 450 °C for 3 h. The compositions of the catalyst were as follows Fe:Mn:K:Si = 100:15:3:30 (molar ratio).

2.2. Catalyst characterizations

MES was tested in a MBF-1100 micro-reactor with Mylar windows, which allows the experiments using different agents such as H_2 , CO or syngas. The catalyst was reduced or carburized in different temperatures (such as 260, 280, 300 and 350 °C) with 60 ml/min of gas flow rate and 1 °C/min of heating rate. After pretreatment the sample was cooled in nitrogen atmosphere to 50 °C and the Mössbauer spectra were recorded. Isomer shift (IS), quadruple splitting (QS) and magnetic hyperfine field (Hhf) were used as the spectral components.

XPS spectra were taken by a VG MiltiLab 2000 system with Al K α (1486.6 eV) as the X-ray source. The C 1s as a reference signal was adjusted to 284.6 eV. All XPS results were obtained from samples treated in situ in the reaction cell and then transferred into the ultra-high vacuum chamber without exposure to air.

DRIFTS spectra were collected using an infrared spectrometer (Equinox 55, Bruker, Germany), equipped with KBr optics and a MCTD316 detector at the liquid nitrogen temperature (77 K). The infrared cell with ZnSe windows was connected to a feed system with a set of stainless steel gas lines, which allows in situ measurements for CO or CO + H_2 adsorption using argon as the flushing gas. In all cases, the gas flows were maintained at 60 ml/min by mass flow control meters. All spectra were recorded with a resolution of 4 cm^{-1} and accumulation of 64 scans.

TPD experiments were performed on an Autochem 2920 characterization system equipped with a thermal conductivity detector (TCD), using Ar (H_2 -TPD) or He (CO-TPD) as the carrier gas. The sample loading was 200 mg. The gas flow rate was 60 ml/min and the heating rate was 1 °C/min. The pretreatment conditions were the same as MES. After pretreatment the sample was cooled to 50 °C for the TPD tests. H_2 or CO adsorption on catalyst was performed at 50 °C for 30 min, and then the sample was purged with Ar or He for 30 min to remove weakly adsorbed species, respectively. After this step, H_2 - or CO-TPD was carried out while the temperature was increased to 800 °C.

3. Results and discussion

3.1. Reduction behavior of iron phases

3.1.1. Phase evolution in the bulk regions

The MES patterns of the catalyst reduced in H_2 atmosphere with different temperatures are shown in Fig. 1. Table 1 lists the corresponding iron-phase compositions in the bulk regions. The Mössbauer spectrum of the fresh catalyst shows one sextet (IS of 0.37 mm/s, QS of 0.19 mm/s and Hhf of 507 kOe) and one double (IS of 0.32 mm/s and QS of 0.82 mm/s), which correspond to $\alpha-Fe_2O_3$ and superparamagnetic (spm) Fe^{3+} ions, respectively. Reduction at 260 °C for 12 h results in the formation of Fe_3O_4 and (spm) Fe^{2+} . The values of the doubles with IS of 0.80–0.82 mm/s and QS of 1.10–1.23 mm/s for (spm) Fe^{2+} are attributed to the formation of metastable FeO [27]. The appearance of metastable FeO during

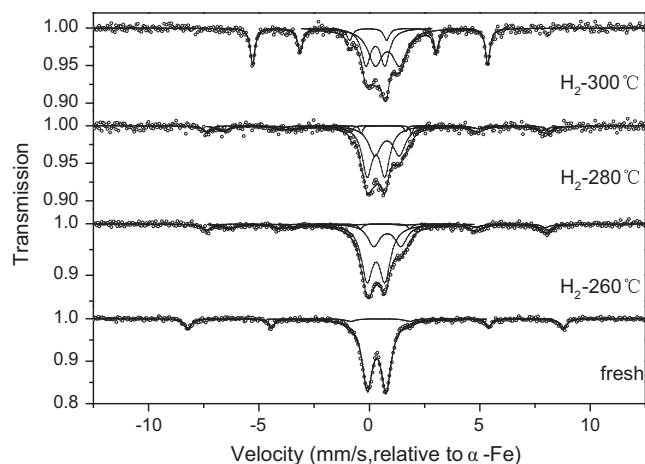


Fig. 1. Mössbauer spectra of the H_2 -reduced catalysts at different temperatures for 12 h.

reduction may be ascribed to the interaction of metal-support in the catalyst, remaining the stable state of metastable FeO [16]. With the increase of reduction temperature the amounts of Fe_3O_4 decrease whereas that of FeO increase (shown in Table 1), implying that the reduced Fe_3O_4 is converted gradually to FeO with the proceeding of reduction. As reduction temperature increases continually to 300 °C, there is the formation of large amounts of bulky metallic iron (about 30%). In addition, it can be found from Table 1 that the reduction degree of iron phases presents an increasing trend with increasing temperature. All of these results suggest that phase transformation in the bulk regions for the Fe–Mn supported catalyst in H_2 atmosphere follows the order $\alpha\text{-Fe}_2O_3 \rightarrow Fe_3O_4 \rightarrow FeO \rightarrow \alpha\text{-Fe}$.

3.1.2. Phase evolution on the surface layers

XPS are used to investigate phase evolution on the surface layers. From Fig. 2, it can be found that a Fe $2P_{3/2}$ peak at 710.6 eV is observed on the surface layers of the fresh catalyst, which is ascribed to $\alpha\text{-Fe}_2O_3$. With the proceeding of reduction the Fe $2P_{3/2}$ peak becomes broader and shifts toward lower binding energies, corresponding to the formation of Fe_3O_4 [28], indicating that $\alpha\text{-Fe}_2O_3$ on the surface layers is also reduced firstly to Fe_3O_4 . As reduction temperature increases above 280 °C, two weak Fe $2P_{3/2}$ peaks appeared at 709.2 eV and 706.9 eV are observed, which may be attributed to Fe^{2+} and Fe^0 [29], respectively. This implies that the formed Fe_3O_4 is reduced further to Fe^{2+} and Fe^0 phases

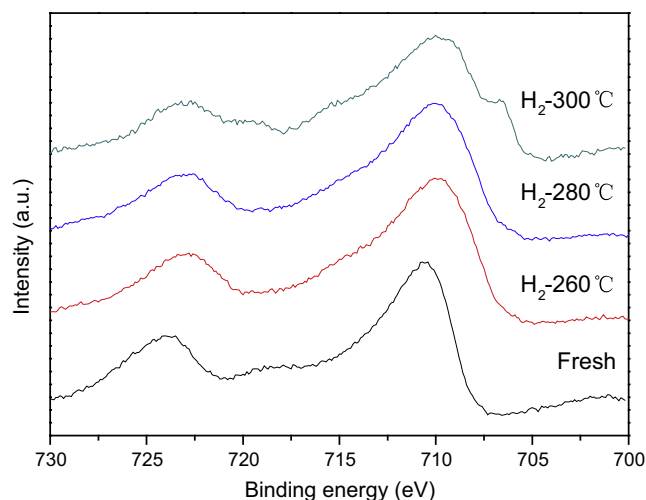


Fig. 2. XPS spectra in Fe 2p region of the H_2 -reduced catalysts at different temperatures for 12 h.

on the surface layers during reduction. With increasing reduction temperature the intensity of Fe^0 species peaks presents an increasing trend, indicating that metallic iron species are formed gradually on the surface layers with the proceeding of reduction. Combined with the results in both the MES and XPS techniques, it is considered that phase transformation in the catalyst reduction with hydrogen involves $\alpha\text{-Fe}_2O_3 \rightarrow Fe_3O_4 \rightarrow FeO \rightarrow \alpha\text{-Fe}$, both occurring in the bulk and on the surface layers.

3.2. Chemisorption behaviors

3.2.1. H_2 -TPD

The chemisorption behaviors of different iron species are investigated by both the H_2 -TPD and CO-TPD techniques. Fig. 3 shows H_2 adsorption behavior on the surface of reduced iron species. A strong desorption peak at 336 °C is observed in H_2 -TPD after reduction at 260 °C. According to the results of MES and XPS, reduction at 260 °C results mainly in the formation of Fe_3O_4 in both the bulk and surface regions. Thus, the peak at 336 °C may be ascribed to the desorption peak of H_2 on Fe_3O_4 . With increasing reduction temperature H_2 adsorption on Fe_3O_4 shifts toward lower temperature, and two new adsorption peaks are appeared at about 150 °C and 100 °C. Salama et al. [30] reported that hydrogen is easily combined with oxygen atoms in oxides to form O–H bonds as H_2 is absorbed on the surface of oxides. The interaction of O–H bonds

Table 1

Mössbauer parameters of the H_2 -reduced catalysts at different temperatures for 12 h.

Catalyst	Phases	Mössbauer parameters				Degree of reduction (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	Area (%)	
Fresh	$\alpha\text{-Fe}_2O_3$	0.37	0.19	507	17.5	
	Fe^{3+}	0.32	0.82		82.5	
H_2 -260	Fe_3O_4	0.36	0.03	481	10.2	10.50
		0.73	0.03	441	11.1	
	Fe^{3+}	0.30	0.81		54.3	
	Fe^{2+}	0.81	1.23		24.4	
H_2 -280	Fe_3O_4	0.41	−0.02	483	9.2	13.60
		0.69	−0.17	449	11.8	
	Fe^{3+}	0.30	0.80		45.2	
	Fe^{2+}	0.80	1.11		33.8	
H_2 -300	Fe^0	−0.01	0.09	331	30.0	44.13
	Fe^{3+}	0.27	0.86		27.6	
	Fe^{2+}	0.82	1.10		42.4	

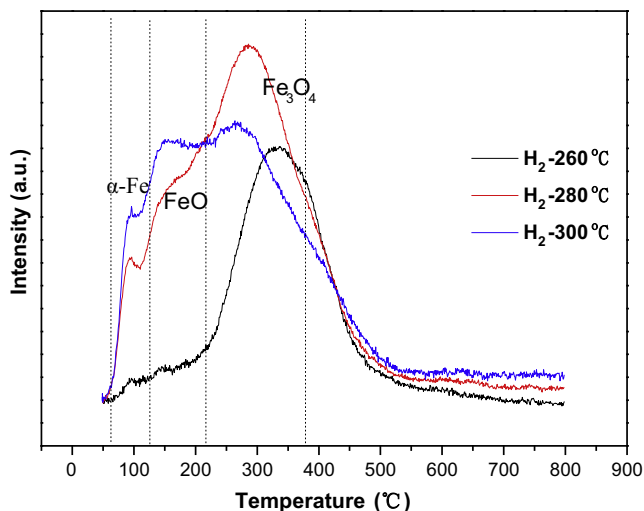


Fig. 3. H₂-TPD of the H₂-reduced catalysts at different temperatures for 12 h.

is strengthened with the increasing amounts of oxygen atoms in oxides, restraining the desorption of H₂ from the oxides. The XPS characterization results indicate that reduction above 280 °C leads to the formation of Fe²⁺ and Fe⁰ on the surface layers. All of these suggest that the peaks of H₂ desorption at about 150 and 100 °C may be attributed to the desorption peaks of hydrogen on FeO and α-Fe, respectively. In addition, from H₂-TPD it can be found that the peak intensity on Fe₃O₄ increases firstly, and then decreases slowly with increasing reduction temperature, suggesting that Fe₃O₄ is formed firstly from α-Fe₂O₃, and then converted gradually to FeO and α-Fe with the proceeding of reduction. The intensity of H₂ desorption peaks on both the FeO and α-Fe presents an increasing trend with increasing temperature, further confirming the conversion of Fe₃O₄ to FeO and metallic iron.

3.2.2. CO-TPD

Fig. 4 shows the adsorption behavior of CO on different iron species. As shown in Fig. 4, there are three desorption peaks of CO at about 200, 280 and 440 °C after reduction at 260 °C. The results mentioned above indicate that reduction in H₂ at 260 °C leads mainly to the formation of Fe(II) oxide species. As CO is adsorbed on the surface of iron oxides, the weakening interaction between CO and oxygen atoms results easily in the desorption of

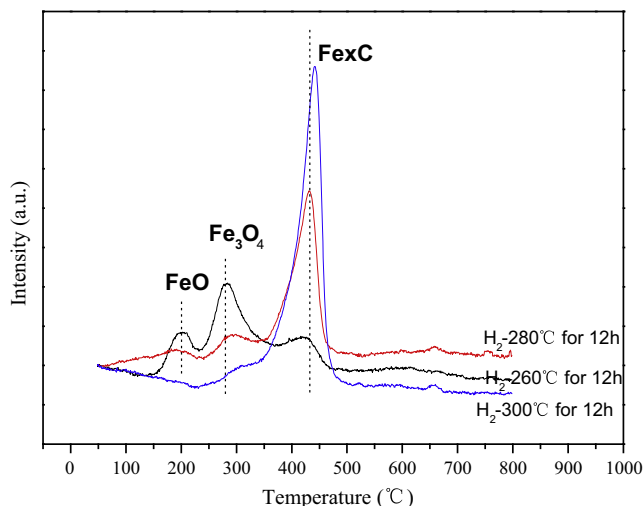


Fig. 4. CO-TPD of the H₂-reduced catalysts at different temperatures for 12 h.

CO from iron oxides [31]. Therefore, the peaks of CO desorption at 200 and 280 °C may be attributed to the weak adsorption of CO on FeO and Fe₃O₄, respectively. In addition, the adsorption peak of CO on metallic iron should be appeared in lower temperature due to the weak combination of CO and metallic iron, whereas no desorption peak of CO on α-Fe is observed in lower temperature in CO-TPD patterns, implying that the formed α-Fe may be converted to iron carbides after adsorption of CO. Furthermore, CO is dissociated easily to C and O atoms in Fischer–Tropsch synthesis, promoting the formation of iron carbides via the combination of C atoms and iron species. Thus, the adsorption peak of CO at higher temperature (440 °C) may be attributed to the adsorption of CO on iron carbides. It is apparent that the intensity of CO desorption peak on iron carbides increases continually while that on reduced iron oxides presents a decreasing trend with increasing temperature, indicating that all of reduced iron species convert gradually to iron carbides after adsorption of CO. Accompanied with the gradual reduction of Fe₃O₄ to FeO, the decreasing trend of the intensity of CO adsorption peak on FeO is more obvious than that on Fe₃O₄, implying that carburization ability of FeO is stronger than that of Fe₃O₄. No appearance of CO adsorption peak on α-Fe suggests that α-Fe is more easily converted to iron carbides in contrast to Fe₃O₄ and FeO.

MES is used to further investigate carburization behavior of metallic iron. From Fig. 5 and Table 2 it can be found that large amounts of α-Fe (about 60%) are formed after reduction in hydrogen atmosphere at 350 °C for 12 h. Subsequently, the formed α-Fe is converted completely to iron carbides (especially for χ-Fe₅C₂) after adsorption of CO for only 1 h, while the amounts of Fe(II) oxide species decrease slightly, further validating that α-Fe is the easiest converted to iron carbides in reduced iron species. Therefore, it is considered that carburization ability of reduced iron species is following the order α-Fe > FeO > Fe₃O₄. In addition, the amounts of both the iron carbides and iron oxides remain unchanged basically with carburization time on stream, implying that reduced iron species are converted gradually to iron carbides with the proceeding of carburization, and both the iron carbides and iron oxides (especially for Fe₃O₄) reach ultimately a balance state.

3.3. DRIFTS

3.3.1. CO adsorption on reduced iron species

In order to investigate the relationship between surface adsorption species and reduced iron phases, the micro-structures in different FTS stages are characterized by DRIFTS. Fig. 6 presents

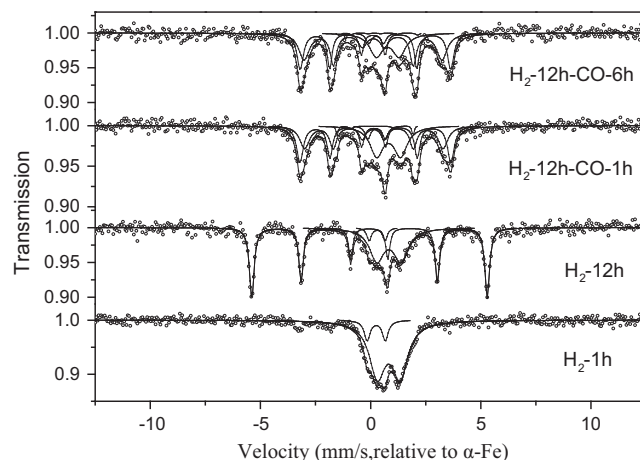
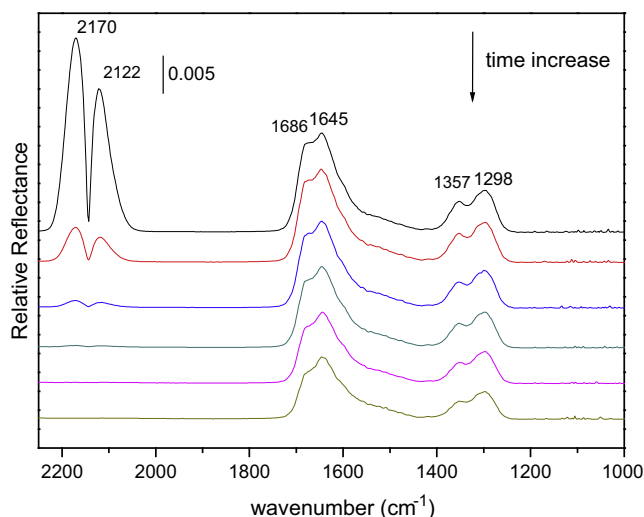


Fig. 5. Mössbauer spectra of the 350 °C H₂-reduced catalyst after carburization for different times in CO atmosphere.

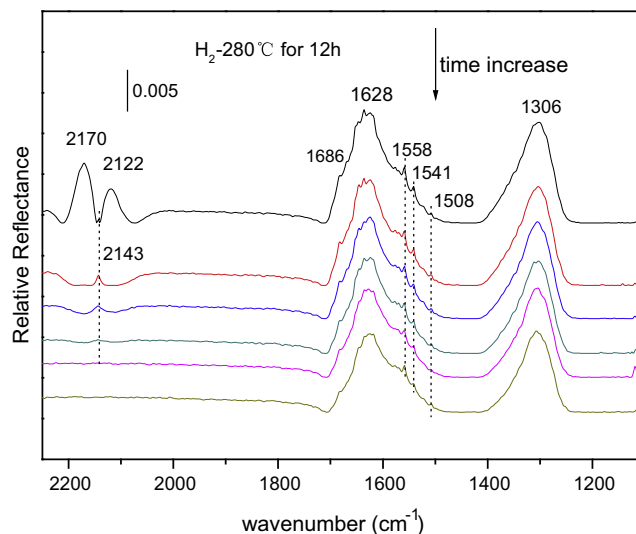
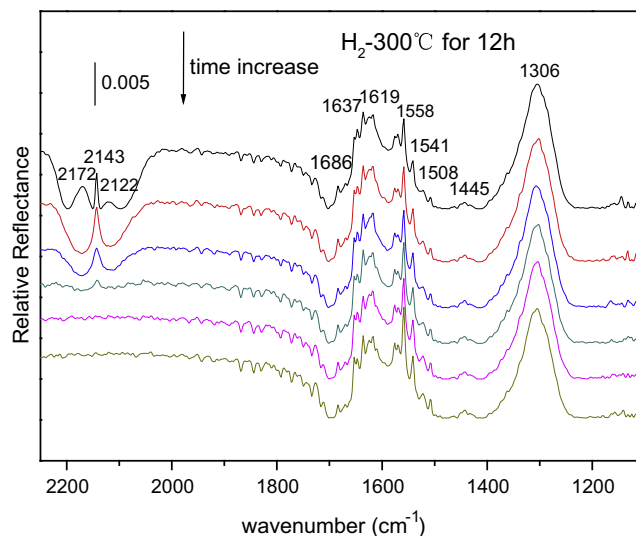
Table 2Mössbauer parameters of the 350 °C H₂-reduced catalyst after carburization for different times in CO atmosphere.

Catalyst	Phases	Mössbauer parameters			
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	Area (%)
H ₂ -1 h	(spm) Fe ³⁺	0.26	0.82		12.5
	(spm) Fe ²⁺	0.82	1.03		87.5
H ₂ -12 h	α -Fe	−0.06	0.01	332	57.0
	(spm) Fe ³⁺	0.29	0.72		4.9
	(spm) Fe ²⁺	0.83	1.09		38.1
	χ -Fe ₅ C ₂	0.18	0.08	213	35.0
H ₂ -12 h-CO-1 h	χ -Fe ₅ C ₂	0.13	0.03	195	31.4
	χ -Fe ₅ C ₂	0.15	0.10	107	2.9
	(spm) Fe ³⁺	0.27	0.80		4.8
	(spm) Fe ²⁺	0.82	1.09		25.9
	χ -Fe ₅ C ₂	0.17	0.10	212	34.0
	χ -Fe ₅ C ₂	0.12	−0.02	195	40.7
H ₂ -12 h-CO-6 h	χ -Fe ₅ C ₂	0.26	0.18	111	1.2
	(spm) Fe ³⁺	0.25	0.67		3.9
	(spm) Fe ²⁺	0.77	1.04		20.1
	χ -Fe ₅ C ₂	0.17	0.10	212	34.0
	χ -Fe ₅ C ₂	0.12	−0.02	195	40.7

**Fig. 6.** Infrared spectra recorded from the adsorption of CO on the fresh catalyst.

infrared spectra of CO on the fresh catalyst. Several bands at 2170, 2120, 1686, 1645, 1357, 1298 cm^{−1} appear on the fresh catalyst after CO adsorption. The doublet bands at 2170 and 2122 cm^{−1} correspond to the vibrations of R- and P-branches of the gaseous CO on the sample [25], which diminish fast upon argon flushing. The bands appeared in the region of 1700–1100 cm^{−1} are assigned to carbonate species formed on the iron oxides. It is apparent that these bands appear as doubles at 1686, 1645 cm^{−1} and 1357, 1298 cm^{−1}, indicating that surface iron species exist in two valence states, corresponding to Fe(III) and Fe(II) oxide species [25]. The formation of Fe(II) oxide species on the fresh catalyst after CO adsorption implies that CO has the strong redox ability to convert part of Fe(III) species to Fe(II) oxides the intensity of these carbonate species remains unchanged basically after argon flushing, suggesting that the adsorption ability of these carbonates on the Fe(III) and Fe(II) oxides is strong.

After reduction in H₂ atmosphere at 260 °C, the catalyst presents the same adsorption species with the fresh catalyst (not shown here), indicating that there are the Fe(III) and Fe(II) oxide species formed on the catalyst reduced at 260 °C. According to the results of XPS and MES mentioned above, reduction at 260 °C results mainly in the formation of Fe₃O₄ and FeO from the bulk to surface layers, which exhibit in forms of Fe(III) and Fe(II) species in infrared spectra. In addition, there is no formation of hydrocar-

**Fig. 7.** Infrared spectra recorded from the adsorption of CO on the 280 °C H₂-reduced catalyst.**Fig. 8.** Infrared spectra recorded from the adsorption of CO on the 300 °C H₂-reduced catalyst.

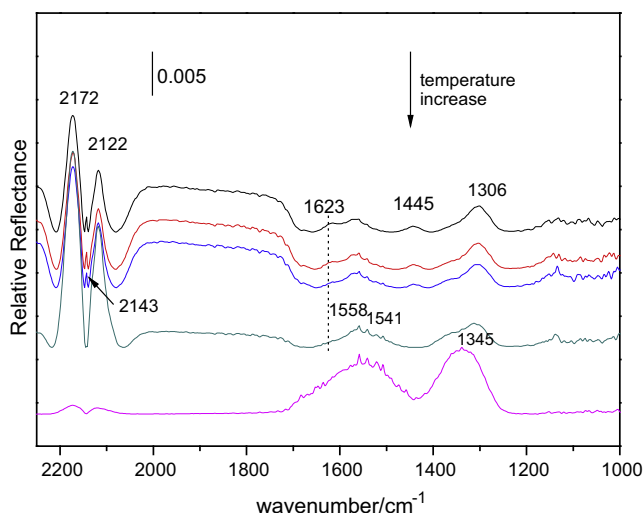


Fig. 9. Infrared spectra recorded from the adsorption of syngas ($H_2/CO = 1.2$) on the 300 °C H_2 -reduced catalyst with increasing temperature.

bons species on the catalyst reduced at 260 °C after CO adsorption, suggesting that the reduced Fe(II) oxide species play possibly a negligible role in providing the active sites for Fischer–Tropsch synthesis.

For the catalyst reduced at 280 °C, some new bands appear in the range of 2200–2000 cm^{-1} and 1700–1100 cm^{-1} (Fig. 7). The band appeared at 2143 cm^{-1} is assigned to CO adsorption on Fe(II) sites, implying that reduction at 280 °C causes the formation of large amounts of Fe(II) sites on the surface layers. The bands at 1686, 1628 and 1306 cm^{-1} are ascribed to carbonates formed on iron oxides, while 1558, 1541 and 1508 cm^{-1} may be attributed to C–H bending modes of the hydrocarbons [21]. The formation of hydrocarbons species on the catalyst suggests that the FTS active sites have been appeared on the surface layers, promoting the formation of hydrocarbons species. The above XPS results show that reduction at 280 °C results in the formation of Fe^{2+} and Fe^0 species.

However, there are not CO adsorption peaks on the Fe^0 species in the range of 2000–1700 cm^{-1} in infrared patterns. Due to the strongest carburization ability of Fe^0 species compared to Fe_3O_4 and FeO, it is possible that the formed Fe^0 species have been carburized completely to iron carbides after CO adsorption. Furthermore, the Fe(II) oxides play a negligible role in promoting the formation of hydrocarbons species. Therefore, the formation of hydrocarbons species on the catalyst may be attributed to conversion of reduced iron species to iron carbides on the surface layers, providing the active sites for Fischer–Tropsch synthesis.

With the increase of reduction temperature, the intensity of adsorbed species on the catalyst takes place obviously changes. From Fig. 8 it is apparent that the intensity of hydrocarbons species on the catalyst reduced at 300 °C increases obviously compared to that on the catalyst reduced at 280 °C, indicating that reduction at 300 °C promotes the formation of more active sites on the catalyst after CO adsorption. According to the results mentioned above, reduction at 300 °C leads mainly to the formation of α -Fe and Fe(II) oxide sites. However, no CO adsorption peaks on α -Fe in the range of 2000–1700 cm^{-1} are still observed in infrared patterns after reduction at 300 °C, implying that carburization of large amounts of metallic iron on the surface layers provides the more FTS active sites, promoting the formation of hydrocarbons species on the catalyst.

3.3.2. Syngas adsorption on iron phases

The catalyst reduced at 300 °C undergoes the FTS reaction in syngas ($H_2/CO = 1.2$) atmosphere. Infrared spectra of adsorbed species on the catalyst in syngas atmosphere with increasing reaction temperature are shown in Fig. 9. It can be clearly seen that the peak intensity of CO adsorbed on Fe(II) oxide sites (2143 cm^{-1}) decreases gradually and disappears ultimately with increasing temperature, indicating that the Fe(II) species on the surface layers are converted gradually to iron carbides in syngas atmosphere. In addition, no diffraction peaks of CO adsorbed on Fe(0) sites are observed with increasing temperature, further validating that the formed Fe(0) sites are converted completely to iron carbides during FTS. All of these results suggest that with the increase of reac-

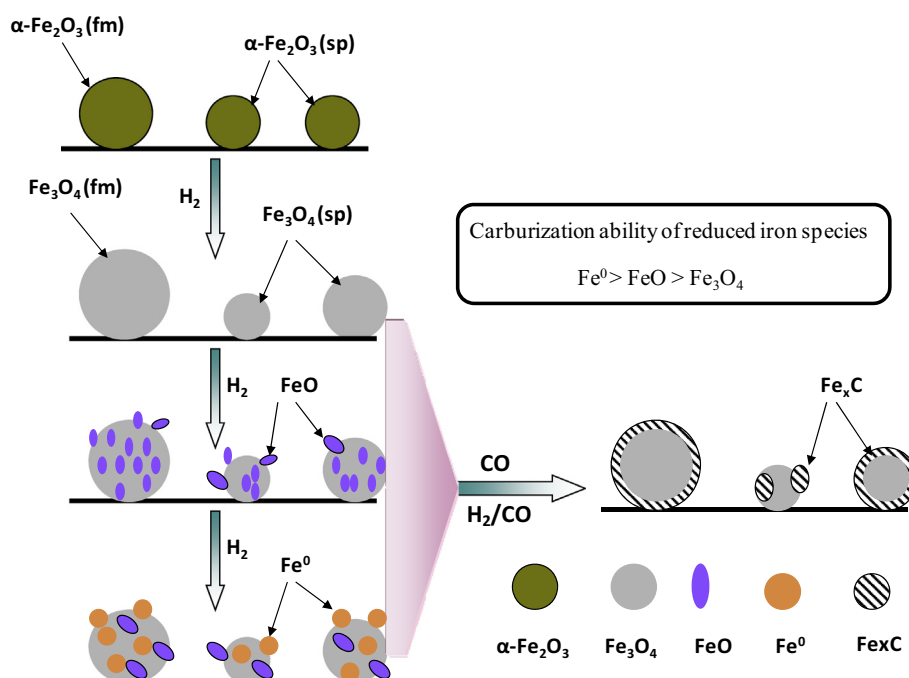


Fig. 10. Carburization behaviors of different reduced iron phases for the iron-based catalyst.

tion temperature in syngas atmosphere the reduced iron phases are carburized mainly to iron carbides on the surface layers. Simultaneously, the amounts of hydrocarbons species absorbed on the catalyst present an increasing trend, further validating that the formation of iron carbides (especially for χ -Fe₅C₂) on the surface layers provides the active sites for FTS. In our previous results [32], the Fe–Mn catalyst was carried out under reaction conditions of 280 °C, 1.50 MPa and H₂/CO = 1.2. The catalytic activity presented an increasing trend with the gradual formation of iron carbides (especially for χ -Fe₅C₂) on the surface layers. Combined with the results in the present study, it suggests that reduced iron species (such as Fe₃O₄, FeO and α -Fe) are carburized to iron carbides on the catalyst, promoting the formation of hydrocarbons species.

3.4. Reduction and carburization behaviors of iron species

Combined with the characterization results in both the bulk and surface regions, a clear scene is obtained on reduction and carburization behaviors of different iron species (shown in Fig. 10). In H₂ atmosphere the α -Fe₂O₃ in the fresh catalyst is firstly reduced to Fe₃O₄ in both the bulk and surface regions. The formed Fe₃O₄ is reduced continually to metastable FeO in the role of metal-support interaction. The metastable FeO is then reduced to α -Fe, both occurring in the bulk and on the surface layers. The reduced iron phases such as Fe₃O₄, FeO and α -Fe can be converted to iron carbides in CO and/or syngas atmosphere. Carburization ability of reduced iron species is following the order α -Fe > FeO > Fe₃O₄. Due to the strongest carburized ability of α -Fe, iron carbides are formed mainly on Fe₃O₄ sites. Ultimately, both the iron carbides and Fe(II) oxide species (especially for Fe₃O₄) reach a balance state during FTS. The redox ability of carbon monoxide may convert part of Fe(III) species to Fe(II) species on the fresh catalyst, resulting to the formation of carbonate species on the surface of both the Fe(III) and Fe(II) oxide sites. With the gradual carburization of reduced iron phases, hydrocarbons species are formed slowly on the catalyst, except for the carbonate species adsorbed on the surface layers. As the surface iron phases are converted completely to iron carbides (especially for χ -Fe₅C₂), the hydrocarbons species are formed mainly on the catalyst, suggesting that iron carbides (especially for χ -Fe₅C₂) formed on the surface layers play a positive role in providing the FTS active sites.

4. Conclusions

In this study, combined methods (MES/XPS, DRIFTS and H₂&CO-TPD) were applied to characterize the reduction and carburization behaviors as well as adsorption properties of different iron phases over a precipitated Fe–Mn catalyst. Phase transformation of iron species in both the bulk and surface regions in H₂ atmosphere was followed as α -Fe₂O₃ → Fe₃O₄ → FeO → α -Fe. The strong interaction of metal-support caused the formation of metastable FeO. All of the reduced iron phases such as Fe₃O₄, FeO and α -Fe could be converted to iron carbides in CO or syngas atmosphere, whereas carburization ability of these reduced iron species was followed the order α -Fe > FeO > Fe₃O₄. Hydrocarbons species were formed mainly on the catalyst as reduced iron phases on the surface layers were carburized completely to iron carbides.

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References

- [1] Rahimpour MR, Bahmanpour AM. Optimization of hydrogen production via coupling of the Fischer–Tropsch synthesis reaction and dehydrogenation of cyclohexane in GTL technology. *Appl Energy* 2011;88:2027–36.
- [2] Ding M, Yang Y, Li Y, Wang T, Ma L, Wu C. Impact of H₂/CO ratios on phase and performance of Mn-modified Fe-based Fischer Tropsch synthesis catalyst. *Appl Energy* 2013;112:1241–6.
- [3] DiGenova KJ, Botros BB, Brisson JG. Method for customizing an organic Rankine cycle to a complex heat source for efficient energy conversion, demonstrated on a Fischer Tropsch plant. *Appl Energy* 2013;102:746–54.
- [4] Galvis HMT, Koeken ACJ, Bitter JH, Davidian T, Ruitenbeek M, Dugulan AI, et al. Effects of sodium and sulfur on catalytic performance of supported iron catalysts for the Fischer–Tropsch synthesis of lower olefins. *J Catal* 2013;303:22–30.
- [5] Yu G, Sun B, Pei Y, Xie S, Yan S, Qiao M, et al. FeOy@C spheres as an excellent catalyst for Fischer–Tropsch synthesis. *J Am Chem Soc* 2010;132:935–7.
- [6] Herranz T, Rojas S, Pérez-Alonso FJ, Ojeda M, Terreros P, Fierro JLG. Genesis of iron carbides and their role in the synthesis of hydrocarbons from synthesis gas. *J Catal* 2006;243(1):199–211.
- [7] Ning W, Koizumi N, Chang H, Mochizuki T, Itoh T, Yamada M. Phase transformation of unpromoted and promoted Fe catalysts and the formation of carbonaceous compounds during Fischer–Tropsch synthesis reaction. *Appl Catal A* 2006;312:35–44.
- [8] Pendyala VRR, Graham UM, Jacobs G, Hamdeh HH, Davis BH. Fischer–Tropsch synthesis: morphology, phase transformation, and carbon-layer growth of iron-based catalysts. *ChemCatChem* 2014;6(7):1952–60.
- [9] Mahajan D, Güttlich P, Stumm U. The role of nano-sized iron particles in slurry phase Fischer–Tropsch synthesis. *Catal Commun* 2003;4:101–7.
- [10] Li S, Ding W, Meitzner GD, Iglesia E. Spectroscopic and transient kinetic studies of site requirements in iron-catalyzed Fischer–Tropsch synthesis. *J Phys Chem B* 2002;106:85–91.
- [11] Cui X, Xu J, Zhang C, Yang Y, Gao P, Wu B, et al. Effect of pretreatment on precipitated Fe–Mo Fischer–Tropsch catalysts: morphology, carburization, and catalytic performance. *J Catal* 2011;282:35–46.
- [12] Chen W, Fan Z, Pan X, Bao X. Effect of confinement in carbon nanotubes on the activity of Fischer–Tropsch iron catalyst. *J Am Chem Soc* 2008;130:9414–9.
- [13] Li S, Krishnamoorthy S, Li A, Meitzner GD, Iglesia E. Promoted iron-based catalysts for the Fischer–Tropsch synthesis: design, synthesis, site densities, and catalytic properties. *J Catal* 2002;206(2):202–17.
- [14] Mannie GJA, Lammich L, Li YW, Niemantsverdriet JW, Lauritsen JV. Monolayer iron carbide films on Au (111) as a Fischer–Tropsch model catalyst. *ACS Catal* 2014;4:3255–60.
- [15] Davis BH. Fischer–Tropsch synthesis: relationship between iron catalyst composition and process variables. *Catal Today* 2003;84:83–98.
- [16] Bukur DB, Lang X, Ding Y. Pretreatment effect studies with a precipitated iron Fischer–Tropsch catalyst in a slurry reactor. *Appl Catal A* 1999;186(1–2):255–75.
- [17] Zhang CH, Yang Y, Teng BT, Li TZ, Zheng HY, Xiang HW, et al. Study of an iron-manganese Fischer–Tropsch synthesis catalyst promoted with copper. *J Catal* 2006;237(2):405–15.
- [18] Kumar A, Miller JT, Mukasyan AS, Wolf EE. In situ XAS and FTIR studies of a multi-component Ni/Fe/Cu catalyst for hydrogen production from ethanol. *Appl Catal A* 2013;467:593–603.
- [19] Rodulfo-Baechler SM, González-Cortés SL, Orozco J, Sagredo V, Fontalé B, Morad AJ, et al. Characterization of modified iron catalysts by X-ray diffraction, infrared spectroscopy, magnetic susceptibility and thermogravimetric analysis. *Mater Lett* 2004;58:2447–50.
- [20] Bian G, Oonuki A, Kobayashi Y, Koizumi N, Yamada M. Syngas adsorption on precipitated iron catalysts reduced by H₂, syngas or CO and on those used for high-pressure FT synthesis by in situ diffuse reflectance FTIR spectroscopy. *Appl Catal A* 2001;219:13–24.
- [21] Boellaard E, van der Kraan AM, Geus JW. Behaviour of a cyanide-derived Fe/Al₂O₃ catalyst during Fischer–Tropsch synthesis. *Appl Catal A* 1996;147(1):229–45.
- [22] Guglielminotti E. Spectroscopic characterization of the Fe/ZrO₂ system: 1. CO adsorption. *J Phys Chem* 1994;98:4884–91.
- [23] Guglielminotti E. Spectroscopic characterization of the Fe/ZrO₂ system. 2. NO adsorption and CO–NO reaction. *J Phys Chem* 1994;98:9033–8.
- [24] Jiang M, Koizumi N, Yamada M. Characterization of potassium-promoted iron-manganese catalysts by in situ diffuse reflectance FTIR using NO, CO and CO + H₂ as probes. *Appl Catal A* 2000;204(1):49–58.
- [25] Jiang M, Koizumi N, Yamada M. Adsorption properties of iron and iron-manganese catalysts investigated by in-situ diffuse reflectance FTIR spectroscopy. *J Phys Chem B* 2000;104:7636–43.
- [26] Yang Y, Xiang HW, Xu YY, Bai L, Li YW. Effect of potassium promoter on precipitated iron-manganese catalyst for Fischer–Tropsch synthesis. *Appl Catal A* 2004;266(2):181–94.
- [27] Xu J, Bartholomew CH. Temperature-Programmed Hydrogenation (TPH) and in situ Mössbauer spectroscopy studies of carbonaceous species on silica-supported iron Fischer–Tropsch catalysts. *J Phys Chem B* 2005;109(6):2392–403.
- [28] Bukur DB, Okabe K, Rosynek MP, Li C, Wang D, Rao KRPM, et al. Activation studies with a precipitated iron catalyst for Fischer–Tropsch synthesis: I. Characterization studies. *J Catal* 1995;155(2):353–65.

- [29] Bonnet F, Ropital F, Lecour P, Espinat D, Huiban Y, Gengembre L, et al. Study of the oxide/carbide transition on iron surfaces during catalytic coke formation. *Surf Interface Anal* 2002;34:418–22.
- [30] Salama TM, Ebitani HK, Hattori H, Kita H. Electron-paramagnetic-resonance studies of reduced platinum supported on titanium-oxide-effect of hydrogen adsorption-desorption on paramagnetic platinum(I) and titanium(III) species. *Chem Mater* 1994;6(1):21–6.
- [31] Qin ZH, Lewandowski M, Sun YN, Shaikhutdinov S, Freund HJ. Encapsulation of Pt nanoparticles as a result of strong metal-support interaction with Fe_3O_4 (111). *J Phys Chem C* 2008;112(27):10209–13.
- [32] Ding M, Yang Y, Wu B, Wang T, Ma L, Xiang H, et al. Transformation of carbonaceous species and its influence on catalytic performance for iron-based Fischer-Tropsch synthesis catalyst. *J Mol Catal A* 2011;351:165–73.